

Screening for regulated plants with Potency Enhancement Properties

A chromatographic Approach

Be aware of herbal adulteration, especially for plant food supplements!

This kind of medicines in disguise, contain undeclared active or medicinal plants, often regulated or forbidden, potentially causing side effects and interaction with existing therapies.

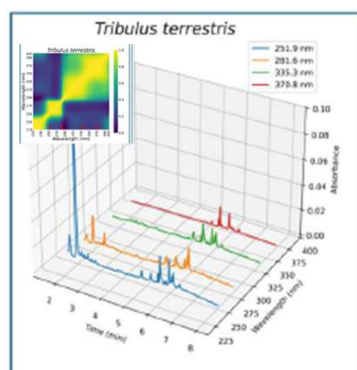
Need for screening approaches!

Problem: Most OMCLs dealing with falsified medicines are chemical laboratories,

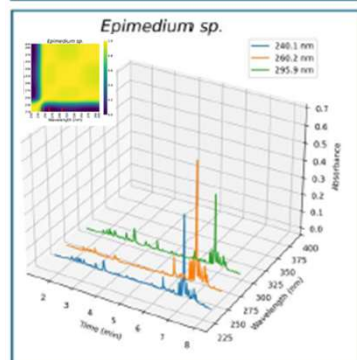
Concept: Chromatographic fingerprinting

Case study: Plant food supplements for potency enhancement

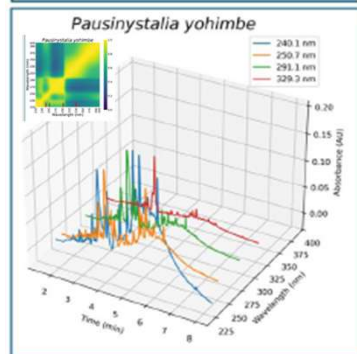
- ✓ One UHPLC –DAD method for three herbal potency enhancers
- ✓ Selection of most characteristic wavelengths per plant (correlation)
- ✓ Binary PLS-DA modelling
- ✓ Screening of 25 suspicious samples



PLS-DA model (+/-)



PLS-DA model (+/-)



PLS-DA model (+/-)

Illegal samples	claimed presence of the plants	Results for Epimedium sp.,	Results for Tribulus terrestris	Results for Pausinystalia yohimbe	Comparison with label claim
1	-	-	-	-	Concordance
2	Tribulus terrestris	-	Absent	Present	Discrepancy
3	Tribulus terrestris	Present	Absent	Present	Discrepancy
4	-	-	-	-	Concordance
5	-	Present	Present	-	Finding
6	Tribulus terrestris	-	Present	-	Concordance
7	Tribulus terrestris	-	Present	Present	Discrepancy
8	-	-	Present	-	Finding
9	-	-	Present	Present	Discrepancy
10	-	-	Present	Present	Discrepancy
11	-	-	Present	-	Finding
12	-	Present	Present	-	Finding
13	-	-	Present	-	Finding
14	-	-	Present	-	Finding
15	-	-	Present	-	Finding
16	-	-	Present	-	Finding
17	-	-	Present	-	Finding
18	-	-	-	-	Concordance
19	Tribulus terrestris	-	Present	-	Concordance
20	Tribulus terrestris	Present	Present	-	Discrepancy
21	-	-	-	-	Concordance
22	Epimedium sp.	Absent	Present	-	Discrepancy
23	-	Present	Present	-	Finding
24	Pausinystalia yohimbe	-	Present	Absent	Discrepancy
25	-	Present	-	Present	Finding

76% positif for fraud and/or adulteration

Conclusion: « Medicines in disguise » a wider problem than illegal APIs!



Eric Deconinck and Surbhi Ranjan
 Contact: Eric Deconinck • T. +32 642 51 70 •
 eric.deconinck@sciensano.be

